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Combining superionic conduction and favorable decomposition products in the crystalline lithium-boron-sulfur system: a new mechanism for stabilizing solid Li-ion electrolytes

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Abstract

We report a solid-state Li-ion electrolyte predicted to exhibit simultaneously fast ionic conductivity, wide electrochemical stability, low cost, and low mass density. We report exceptional DFT-based room temperature single crystal ionic conductivity values for two phases within the crystalline lithium-boron-sulfur (Li-B-S) system: 62 (+9, -2) mS cm⁻¹ in Li₅B₇S₁₃ and 80 (-56, -41) mS cm⁻¹ in Li₉B₁₉S₃₃. We report significant ionic conductivity values for two additional phases: between 0.0056 - 0.16 mS/cm mS cm⁻¹ in Li₂B₂S₅, and between 0.0031- 9.7

mS cm⁻¹ in Li₃BS₃ depending on the room temperature extrapolation scheme used. To our knowledge, our prediction gives Li₉B₁₉S₃₃ and Li₅B₇S₁₃ the second- and third-highest reported DFT-computed single crystal ionic conductivities of any crystalline material. We compute the thermodynamic electrochemical stability window widths of these materials to be: 0.50 V for Li₅B₇S₁₃, 0.16 V Li₂B₂S₅, 0.45 V for Li₃BS₃, and 0.60 V for Li₉B₁₉S₃₃. Individually, these materials exhibit similar or better ionic conductivity and electrochemical stability than the best known sulfide-based solid-state Li-ion electrolyte materials, including Li₁₀GeP₂S₁₂. However, we predict that electrolyte materials synthesized from a range of compositions in Li-B-S system may exhibit even wider thermodynamic electrochemical stability windows of 0.63 V and possibly as high as 3 V or greater. The Li-B-S system also has low elemental cost of approximately 0.05 USD/m² per 10 μm thickness, significantly lower than that of germanium-containing LGPS, and a comparable mass density below 2 g/cc. These fast conducting phases were initially brought to our attention by a machine learning-based approach to screen over 12,000 solid electrolyte candidates, and the evidence provided here represents an inspiring success for this model.

Keywords: solid ion conductor; solid electrolyte; Li-ion battery; density functional theory; ion transport; electrochemical stability; kinetics

Introduction

All-solid-state Li-ion batteries (SSLIBs) hold promise as safer, longer lasting, and more energy dense alternatives to today's commercialized LIBs with liquid electrolytes. However, the discovery and design of suitable solid electrolyte materials for use in SSLIBs remains a significant engineering challenge. A high performance solid electrolyte must simultaneously exhibit fast Li ion conduction, negligible electronic conduction, and robust electrochemical stability. The material should also be made from cheap and low mass elements in order to ensure cost competitiveness and high energy density. Ceramic materials are an attractive class of materials for this application,^[1] but all ceramic lithium electrolyte materials heretofore reported in the literature falter on at least one of these critical properties.^[2] We report here evidence that solid electrolytes from the lithium-boron-sulfur (Li-B-S) chemical system display promising predicted values for all these properties, comparable to or exceeding those of the best known ceramic electrolyte materials.

A common discovery strategy for new solid electrolyte materials is to search chemical space for a material that simultaneously possesses fast ionic conduction and a wide electrochemical stability window. This strategy is guided by the empirical observation that electrolytes with an electrochemical stability window narrower than the potential difference across the electrodes undergo degradation reactions at the electrode-electrolyte interface (EEI) that result in interfacial products with typically poor ionic conductivity.^[3] These interfacial phases then block ionic motion between electrodes and halt battery operation. Typically, degradation passivates the electrode against further reaction, but degradation can continue until the entire electrolyte reacts if one or more of the EEI degradation products are electron conductors.

The material LGPS ($\text{Li}_{10}\text{GeP}_2\text{S}_{12}$) is currently considered among the best available lithium ion conductors, with an ionic conductivity of 12 mS cm^{-1} at room temperature.^[4] Unfortunately, LGPS also exhibits a very narrow (thermodynamic) electrochemical stability window of approximately 0.3 V (2.1 – 2.4 V vs. Li/Li^+).^[5] Computational and experimental studies have been reported that suggest metallic and ionically insulating reaction products form on the lithium metal anode interface when operated outside of this electrochemical stability window.^[5,6] Typical commercial Li-ion batteries exhibit a voltage difference of approximately 4 V across cathode and anode (approx. 3.8 V – 0.1 V vs. Li/Li^+ for a transition metal oxide/graphite cell),^[7,8] and thus LGPS can only withstand a small fraction of the voltage range required for significant energy density.

Our recent work^[9] and work by Muy et al.^[10] suggests that ionic conductivity and electrochemical stability may be inversely correlated, suggesting that optimal materials are outliers and likely to be very difficult to find. More creative solutions may be required to circumvent this limitation. We propose an alternative approach to the strategy of searching for one material that simultaneously optimizes both criteria: identifying a high ionic conductivity solid material that breaks down into fast ion conducting and electronically insulating phases at both EEs. In this scenario, a wider effective electrochemical window may be achieved because the electrolyte passes through more phase transitions before poor ion conductors actually appear. Realizing this strategy is an exercise in materials discovery: no existing materials appear to have this property, so new materials must be discovered.

Here we predict crystalline Li-B-S is a promising material system that may demonstrate this capability: we find evidence that phase mixtures of fast ion conducting phases from the Li-B-S system, $\text{Li}_5\text{B}_7\text{S}_{13}$, $\text{Li}_2\text{B}_2\text{S}_5$, Li_3BS_3 , and $\text{Li}_9\text{B}_{19}\text{S}_{33}$, over a range of boron-to-sulfur ratios will oxidize and reduce into ionically conducting and electronically insulating interfacial products, enabling electrochemical stability over a wider potential range than any of the individual phases alone.

Lithium ion conducting glasses from the B_2S_3 - Li_2S system were discovered in the early 1980s and reported to have a room temperature lithium ion conductivity of $10^{-4} \text{ S cm}^{-1}$.^[11] Doping with LiI was reported to increase the lithium conductivity by an order of magnitude, but the oxidation of iodine resulted in too narrow of an electrochemical stability window for use in SSLIBs.^[12,13] Sulfide-based glassy conductors also tend to be quite hygroscopic, which can present stability and performance issues in battery applications.^[14] The crystalline phases of the Li-B-S system and their solid solutions were first studied with Li-NMR in the 1990s, including their ionic transport properties, but we find no reports of ionic conductivity or electrochemical stability in these works.^{15–17}

We reinvestigate the Li-B-S system with today's *ab initio* simulation tools and materials databases at our disposal, and find new evidence to suggest these materials may be among the most promising solid Li-ion electrolyte materials. We report here evidence that four phases in the crystalline Li-B-S system exhibit high room temperature bulk ionic conductivity ($\sim 10^{-2} \text{ S cm}^{-1}$), without the addition of iodine. Furthermore, a study of the thermodynamics of the phase diagram suggests that, unlike any other known superionic lithium conductors, materials in the Li-B-S

system may exhibit the valuable property of decomposing from one solid solution of superionic conductors to another upon oxidation and reduction. This suggests an electrochemical stability window wider than that of a single sulfide phase may be realizable. Additionally, the low molecular weight and materials cost of Li-B-S materials may make them all the more attractive. We investigate the properties of this system below.

Ionic conductivity

First, we perform an evaluation of the ionic conductivity of four phases in the Li-B-S system: $\text{Li}_5\text{B}_7\text{S}_{13}$, Li_3BS_3 , $\text{Li}_2\text{B}_2\text{S}_5$, and $\text{Li}_9\text{B}_{19}\text{S}_{33}$. We report that all four of these thermodynamically stable phases exhibit high lithium ion conduction at room temperature. These structures were initially characterized in the early 1990s as crystalline phases observed from Li-B-S glasses and have space groups Cc, Pnma, Cmcm, and C2/c respectively.^[18–24] These phases were brought to our attention as potential Li-ion conductors by our machine learning-based superionic predictor model developed in previous work.^[25,26] $\text{Li}_5\text{B}_7\text{S}_{13}$, Li_3BS_3 , $\text{Li}_2\text{B}_2\text{S}_5$ and $\text{Li}_9\text{B}_{19}\text{S}_{33}$ possess generally favorable values for the five structural features used by our machine learned model to predict fast ion conduction: many lithium-lithium neighbors (2.7, 6.0, 4.0, 1.4 respectively, compared to a median value of 2.6 across over 10,000 Li containing materials with characterized structures), small electronegativity difference between sublattice atoms (0.5, 0.9, 0.4, 0.4; median of 0.9), low anion framework coordination numbers (7.7, 7.3, 5.2, 8.4; median of 10), and large lithium-anion separation distance (2.5, 2.5, 2.6, 2.4 Å; median of 2.0 Å). The fifth feature, the lithium-lithium separation distance, is relatively large (3.7, 3.2, 3.3, 3.6 Å; median of 3.2) but this is compensated by the favorable values of the other four features. With these feature values, the machine learning-based model predicts these materials to exhibit superionic lithium

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3 conduction. Although some knowledge in the literature about ion transport in these materials pre-
4 dated our machine learning model, no information about the structure or conductivity of any Li-
5 B-S materials was included in the model. This makes the model's predictions of fast ion
6 conductivity in these phases true generalizations of the model. The computational observations
7 of fast ion conduction we provide here suggest that all four predictions by the model may be
8 correct, representing an exciting validation of our machine learning-based approach.
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19 To predict the room temperature (RT) ionic conductivity in $\text{Li}_5\text{B}_7\text{S}_{13}$, $\text{Li}_2\text{B}_2\text{S}_5$, Li_3BS_3 , and
20 $\text{Li}_9\text{B}_{19}\text{S}_{33}$, we perform density functional theory molecular dynamics (DFT-MD) on single
21 crystals of these materials at 900 K, 700 K, 400 K, and 293 K. We leverage the Perdew-Burke-
22 Ernzerhof (PBE)^[27] methodology of the generalized gradient approximation and the projector
23 augmented wave (PAW) method.^[28] Ionic diffusion in solids is generally not isotropic; this three-
24 dimensional diffusivity value is equivalent to the mean of the diagonal elements of the diffusion
25 tensor, or one-third of the trace. We refer the reader to the Methods section for discussion of the
26 simulation details.
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40 In Figure 1 we plot the logarithm of ionic conductivity versus inverse temperature, and confirm
41 significant Li conduction in all four materials at RT when Arrhenius dynamics are assumed. The
42 RT Li conductivity of $\text{Li}_9\text{B}_{19}\text{S}_{33}$ and $\text{Li}_5\text{B}_7\text{S}_{13}$ in particular are remarkable: 80 (-56, -41) mS cm^{-1}
43 and 62 (+9, -2) mS cm^{-1} , respectively. These are both several times higher than the reported DFT
44 single crystal RT ionic conductivity for LGPS (9 mS cm^{-1}),^[29] and the experimentally measured
45 RT ionic conductivity of polycrystalline LGPS (12 mS cm^{-1}).^[4] The ionic conductivities of the
46 other materials vary significantly depending on the details of the RT extrapolation: when the
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3 results from the 400 K and 293 K simulations are removed from the extrapolations due to
4 uncertain convergence,^[30] the following RT ionic conductivities are calculated: 0.0056 (-0.0052,
5 -0.0056) mS cm⁻¹ for Li₂B₂S₅ and 0.0031 (+0.019, -0.0023) mS cm⁻¹ for Li₃BS₃. When the
6 results of the 400 K simulations are included, the following RT ionic conductivities are
7 calculated: 0.16 (-0.05, -0.14) mS cm⁻¹ for Li₂B₂S₅ and 9.7 (+4, -8) mS cm⁻¹ for Li₃BS₃. To
8 compute uncertainties in these values, we extrapolate down to RT along the upper and lower
9 limits of the high temperature uncertainties. We refer the reader to the Methods section for
10 further discussion of calculating the uncertainties.
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24 These predictions for single crystals are among the highest DFT-predicted Li-ion conductivities
25 in crystalline materials. We note that future experimental studies of ionic conductivity in these
26 materials may be affected by defects, grain boundary resistance, and microstructure, none of
27 which are incorporated into single crystal DFT calculations. The impact of such effects on the
28 overall conductivity is generally not predictable.
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38 We compute the electronic band gaps of Li₅B₇S₁₃, Li₂B₂S₅, Li₃BS₃, Li₉B₁₉S₃₃ with PBE DFT to
39 be 3.6, 2.4, 3.1, and 2.9 eV, respectively. Since PBE DFT simulations tend to underestimate the
40 true band gap by a factor of two or more,^[31] we also compute the band gaps with the more
41 accurate HSE06 hybrid functional^[32] with a single *k*-point; we report values of 4.7, 3.5, 4.2, and
42 4.0 eV, respectively. A high band gap in solid electrolyte materials is desired in order to
43 minimize electrical conductivity and maximize electrochemical stability. Although the electrical
44 conductivity depends on both the electronic density of states and the band gap, calculations for
45 pure Si demonstrate that a true band gap of 1 eV or higher leads to acceptably small levels of
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electron conduction.^[25] Assuming a minimum acceptable band gap of 1 eV, our calculations predict that the band gaps are sufficiently large for the Li-B-S phases to be useful in solid electrolyte applications. Furthermore, the band gap (in eV) is an upper bound on the thermodynamic electrochemical stability window width (in V)^[33,34] so larger band gaps may indicate more robust electrochemical stability. To execute these calculations we leverage the Automatic-FLOW (AFLOW) for Materials Discovery *k*-point path generator^[35,36] and the P4VASP visualization tool. More details on these calculations are provided in the Methods section.

Reactions at the electrode-electrolyte interface

The ternary Li-B-S phase diagram with thermodynamically stable phases is provided in Figure 2. We note the four fast conducting phases discovered here are the only predicted stable phases on the interior of the phase diagram.

Knowing that there are four excellent lithium ion conductors in the Li-B-S chemical space, the next step is to identify the optimal composition and operating range for use as an electrolyte. This requires predicting the phases that will form at arbitrary composition $x\text{Li}+y\text{B}+z\text{S}$, as well as which phases will result when the electrolyte is oxidized/reduced by the cathode/anode at the EEI. Existing methods for computationally predicting the electrochemical stability of electrolyte materials rely exclusively on thermodynamic considerations.^[5,37,38] This approach is valuable for predicting possible interfacial behavior but neglects potentially important kinetics. We first examine thermodynamics and then return to kinetics in the following section.

The thermodynamic approach to predicting electrochemical stability entails evaluating the grand potential function Φ for all phases at a given applied Li chemical potential μ_{Li} and computing the grand potential convex hull. Phases on the convex hull of the grand potential function are thermodynamically electrochemically stable against the applied potential. Since electrodes operate by inserting or removing Li from the electrolyte, applying an electrode at a given potential (in V vs. Li/Li⁺) is equivalent to applying a lithium chemical potential of opposite sign (in eV/atom). The grand potential function is computed as:

$$\Phi = \frac{E - \mu_{\text{Li}}N_{\text{Li}}}{\sum_{i \neq \text{Li}} N_i} \quad (1)$$

E is the standard formation enthalpy of the phase per formula unit, μ_{Li} is the (negative) applied chemical potential in eV, N_{Li} is the number of Li atoms in the phase per formula unit, and $\sum_{i \neq \text{Li}} N_i$ is the number of non-Li atoms per formula unit. For simplicity, we assume the entropic contribution to enthalpy is negligible and replace the enthalpy with the DFT-computed standard formation energy. A survey of experimental thermochemical data of over 30 solid sulfide materials^[39] reveals that the contribution of entropic effects to the formation enthalpy at room temperature is typically around 2% and rarely higher than 7%. In diboron trisulfide (B₂S₃), the contribution is 1.9%. This suggests that neglecting to consider entropy may introduce an uncertainty on the order of tens of meV per atom in our grand potential calculations.

We compute the grand potential function for all known stable crystalline phases of the Li-B-S system from Figure 2 over a range of applied potentials from 0 – 5 V vs. Li/Li⁺. At each applied potential, we construct the convex hull by identifying the phases with lowest grand potential function, and then compute the energy above the convex hull of all phases. The predicted electrochemical stability window of an individual phase is given by the range of potentials for

which the structure lies on the convex hull. In other words, a phase is predicted to be thermodynamically electrochemically stable at a given voltage if its grand potential function is more negative than the grand potential function of a linear combination of other phases, provided the other phases and their relative concentrations preserve the elemental ratios of the original phase.

In the thermodynamic picture, it is assumed that a phase will immediately decompose once it is no longer on the grand potential convex hull. In reality, the system will need to overcome an activation barrier for this reaction to proceed. The thermodynamic reduction and oxidation potentials are therefore typically upper- and lower bounds, respectively, as kinetics may stabilize the original phase at voltages outside of the thermodynamic electrochemical stability window.

Considering only thermodynamics, we calculate the four fast conducting Li-B-S phases to have the following electrochemical stability windows: 0.16 V for $\text{Li}_2\text{B}_2\text{S}_5$, 0.45 V for Li_3BS_3 , 0.50 V for $\text{Li}_5\text{B}_7\text{S}_{13}$, and 0.60 for $\text{Li}_9\text{B}_{19}\text{S}_{33}$. See Methods section for computational details on the construction of the convex hull.

The predicted products resulting from oxidation on the cathode or reduction on the anode depend on the other stable phases present in the phase diagram and their computed grand potential function values. The presence of multiple stable superionic phases within the Li-B-S phase diagram suggests that superionic phases may break down into other superionic phases at certain applied voltages. If the products resulting from oxidation or reduction are other fast conducting

phases, the ionic impedance at the EEs should remain low, and the effective electrochemical window of an electrolyte is widened.

In choosing the best Li-B-S electrolyte composition, it is therefore important to consider the products of redox chemistry at the electrode interfaces. The composition that maximizes both ionic conductivity and electrochemical stability may not necessarily be one of these four pure phases, but may consist of a phase mixture of these phases. To that end, we examine the performance of phase mixtures of these fast conducting structures over a range of compositions. Although we do not know *a priori*, a phase mixture electrolyte may exhibit ion conducting phases over a wider electrochemical range than the pure phases alone. We investigate the predicted interfacial chemistry of these compositions below.

Electrochemical stability of Li-B-S mixture

Using the methods described above, we predict the interfacial chemistry and therefore the effective electrochemical stability windows for all candidate compositions within the Li-B-S space. In a closed system, the Gibbs phase rule predicts up to three phases can coexist in a ternary chemical space; but when a Li chemical potential is introduced this decreases to two.^[40] The 1-2 phases predicted to be thermodynamically stable at a given potential and their molar fractions depend on the ratio of boron to sulfur in the composition. In Figure 3, we provide the molar fraction of thermodynamically stable superionic phases as a function of applied potential and boron molar ratio, $B/(B+S)$. The optimal compositions are those for which the largest molar fraction of superionic phases is stable across the broadest voltage range.

The results of the analysis in Figure 3 show the remarkable result that a high concentration of superionic phases are predicted to be stable over a broad range of compositions and applied potentials in the Li-B-S space. This suggests that superionic mixtures of Li-B-S phases may be able to withstand a range of electrode potentials that would seek to induce redox reactions (variation along x -axis in Figure 3), as well as a range of corrosive chemical conditions on the electrode that would change the composition with respect to B or S (variation along y -axis in Figure 3). For example, high concentration of superionic phases is predicted over a relatively wide range of potentials (1.59 to 2.19 V vs. Li/Li⁺) when the composition contains a B molar fraction of between 0.29 to 0.33; additionally, a high concentration of superionic phases is predicted to be present between 1.59 and 2.22 V vs. Li/Li⁺ between a B molar fraction of 0.37 and 0.4. The full list of the predicted 1-2 thermodynamically stable phases as a function of electrode potential and boron-to-sulfur ratio is provided in Table 1, with superionic phases in bold. While the phases in Table 1 are predicted to form a stable mixture over the given potential ranges, our analysis does not consider possible interfacial resistance between phases in the mixture.

The overall reactions provided in Table 1 consider only thermodynamic energy predictions and do not consider kinetic factors. The degree to which diffusion kinetics affect reaction rates at the solid-solid EEI interface is still largely an open question. All of the reactions predicted in Table 1 result in products with fundamentally different stoichiometries, thus requiring significant diffusion of Li, B, and S, possibly over the nanometer length scale or more. Although Li is predicted to be highly mobile, B and S are not. In fact, our DFT-MD simulations of all four superionic Li-B-S phases show zero migration of B or S in any of these phases over hundreds of

picoseconds of high temperature simulation, in contrast to the significant Li migration we observe. We would expect to see some B or S diffusion in these simulations if the energy barrier to long-range B or S diffusion was small enough for such diffusion to be facile. The absence of any B or S diffusion at all on these simulation timescales suggests that the diffusion barrier for B and S is significant, and we assume likely larger than 0.4 eV, an approximate upper bound ion conduction energy barrier in reasonably conductive materials. Thus, with B and S apparently quite immobile, it is possible the fast ion conducting phases in the starting phase mixture will remain kinetically stabilized at voltages outside of the thermodynamic electrochemical stability windows. This will have the fortunate effect of widening the overall electrochemical stability window of the superionic phases.

As an example of this effect, consider the reaction trajectory for $\text{Li}_2\text{B}_2\text{S}_5$, where $\text{B}/(\text{B}+\text{S}) = 0.286$. The single phase $\text{Li}_2\text{B}_2\text{S}_5$ is predicted to be thermodynamically electrochemically stable between electrode potentials of 2.03V and 2.19V. As the electrode potential rises above 2.19V, the predicted oxidation reaction is $\text{Li}_2\text{B}_2\text{S}_5 \rightleftharpoons 0.25\text{LiS}_4 + 2\text{BS}_2 + 1.75\text{Li}$. Here, LiS_4 is a molecular solid that is predicted to be stable at high potentials based on its computed grand potential function. As the potential falls below 2.03V, the predicted reduction reaction is $\text{Li}_2\text{B}_2\text{S}_5 + 2\text{Li} \rightleftharpoons 1.125\text{Li}_3\text{BS}_3 + 0.125\text{Li}_5\text{B}_7\text{S}_{13}$, replacing one superionic conductor with two others. Below 1.62V, the electrolyte is predicted to reduce further through the following reaction: $1.125\text{Li}_3\text{BS}_3 + 0.125\text{Li}_5\text{B}_7\text{S}_{13} + \text{Li} \rightleftharpoons 1.667\text{Li}_3\text{BS}_3 + 0.333\text{B}$. Below 1.59V, the electrolyte is predicted to degrade further: $1.667\text{Li}_3\text{BS}_3 + 0.333\text{B} + 5\text{Li} \rightleftharpoons 5\text{Li}_2\text{S} + 2\text{B}$. Considering the whole reaction trajectory, at least one superionic Li-B-S phase is predicted to be stable across a window of 0.6 V, i.e. between 1.59 V and 2.19 V.

As the electrode potential decreases below 1.59 V or above 2.19 V, the grand potential of the superionic phases increases linearly with applied potential and they lift off the convex hull to be replaced by ionic insulators. If this degradation does occur, all fast ion conductors will give rise to slow ion conductors and battery operation may cease. However, if we assume these reactants are kinetically stabilized until the grand potential difference between reactants and products exceeds the assumed B and S migration barriers of 0.4 eV/atom, lower reduction potentials and higher oxidation potentials may be reached before degradation. In Figure 4, we plot the grand potential energy above the hull as a function of applied potential for every phase. These calculations suggest that the superionic Li-B-S phases may tolerate potentials as low as approximately 1V before kinetic barriers to reduction are overcome. Similarly, superionic Li-B-S phases may tolerate oxidation potentials as high as 4V before oxidation is kinetically favored, with $\text{Li}_9\text{B}_{19}\text{S}_{33}$ appearing to be particularly stable at high potentials. This suggests that kinetic factors may widen the effective electrochemical stability window of superionic Li-B-S mixtures fivefold, from 0.6V to 3V or more. For context, several known liquid electrolytes are stable in the 1-5 V range and beyond.^[34,41–43] An additional strategy for realizing a wider potential window if necessary might be to combine this electrolyte with degradation resistant materials in a two- or three-electrolyte architecture.^[9]

A key question for the cathodic stability of the Li-B-S system is which phase of elemental boron will form, as Li_2S and elemental boron are predicted to form on the anode between 0.36 and 1.59 V regardless of the B molar ratio. The ground state structure of elemental boron is a question that has received considerable attention in recent years.^[44–46] The PBE DFT calculations in the

Materials Project database predict the ground state to be a semiconductor with predicted band gap of 1.4 eV, but DFT also predicts metallic phases to be kinetically accessible by as little as ~ 0.1 eV per atom. It is also possible that metallic Li-B alloys will form on the anode below 1.59 V, analogous to the metallic Li-Ge alloys that have been reported to form on the lithium metal anode in an LGPS battery.^[6] If metallic phases form, the electron insulating criteria outlined above will be violated and the interfacial products will not passivate the surface from further reaction.

At lower potentials (0-0.36 V), the thermodynamically favored products are Li_2S and metallic LiB.^[47] Since LiB is metallic it also will not passivate the anode from further reaction, suggesting the Li-B-S electrolyte may be unsuitable for use in solid-state batteries with Li metal anodes unless the superionic phases can be kinetically stabilized at these low potentials.

Currently there are no other structures in the ICSD and MP databases in the relevant areas of the Li-B-S phase diagram, and we can find no other structures reported in the literature, but the discovery and/or inclusion of new phases may alter the result of the calculations here. The accuracy of these electrochemical predictions are subject to the accuracy of the DFT energy calculations and the thermodynamic assumptions employed here, as well as the completeness of the structures in the phase diagram. This analysis also does not consider the chemical stability of the Li-B-S phases with respect to electrode materials. The degree of chemical stability depends on the choice of electrodes, and chemical reactivity may produce different decomposition products on the EEIs than the products of electrochemical reactivity.

Energy density and cost

Energy density and cost are important practical considerations for any new battery material and should be incorporated into the materials discovery process. This is especially true for candidate SSLIB chemistries, since there must be potential for significant improvement over the incumbent liquid electrolyte technology to become commercially attractive. In order to improve gravimetric energy density, solid electrolyte materials should be made of low mass elements. In this respect, lithium, boron, and sulfur are favorable elements. The densities of Li_3BS_3 , $\text{Li}_5\text{B}_7\text{S}_{13}$, $\text{Li}_2\text{B}_2\text{S}_5$, and $\text{Li}_9\text{B}_{19}\text{S}_{33}$ are 1.65, 2.06, 1.77, and 2.18 g cc⁻¹, respectively. This is a factor of two less dense than LLTO ($\text{Li}_{0.5}\text{La}_{0.5}\text{TiO}_3$, 4.68 g cc⁻¹) and LLZO ($\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ 4.99 g cc⁻¹), and similar to that of LGPS (1.98 g cc⁻¹).

At the time of publication, the price to purchase elemental Li, B, and S in large industrial quantities is approximately \$20 per kg, \$5 per kg, and \$1 per kg. At this price point, the raw materials cost of an Li-B-S solid electrolyte will be in the range ~0.05 USD per m² per 10 μm of thickness. A recent analysis by Li et al. proposed that the low cost of S and its high earth abundance makes sulfur-based cathodes one of the few cost competitive routes for electrochemical energy storage compared to incumbent technology on a capacity per dollar metric.^[48] Sulfur-based solid electrolytes are desirable for the same reasons. Additionally, Li-B-S is free of germanium (~\$500 per kg) making the raw materials significantly cheaper than those in LGPS. Manufacturing will bring additional costs, and the magnitude of these costs will vary depending on the economics and scalability of the synthesis routes. Regardless, the cost of the raw materials suggest that Li-B-S is a strong candidate for commercial manufacture at a cost below McCloskey's threshold of 10 USD per m² for lithium ion electrolytes.^[49] This cost target

has since been adopted by the US Department of Energy's Advanced Research Projects Agency – Energy (ARPA-E).^[50]

Conclusions

The identification of solid lithium ion conductors that oxidize and reduce into non-metallic ion conducting products at the electrolyte-electrode interface would be a significant advancement towards realizing an all-solid-state lithium-ion battery with superior energy density to traditional lithium ion batteries. To this end, we computationally investigate the electrolyte performance of the Li-B-S chemical system. We predict four structures in the Li-B-S system will exhibit superionic ionic conductivity (10^{-1} – 10^{-6} S cm⁻¹) at room temperature over a range of applied Li potentials. Thermodynamic and kinetic considerations suggest that fast lithium ion conductors may form or persist on the electrode-electrolyte interfaces during electrolytic oxidation and reduction by the electrodes. Li-B-S phase mixtures are generally predicted to be effectively thermodynamically stable in a window of 1.6-2.2 V vs. Li/Li⁺, a voltage range twice that of LGPS, and may be kinetically stabilized over the 1.0-4.0 V range and beyond. In Figure 5, we plot the performance characteristics of known Li-ion conductors and the characteristics of the Li-B-S system we predict here. Taking ionic conductivity as indicative of power density and electrochemical stability window as indicative of energy density, this plot shows that Li-B-S may offer an increase in both energy density and power density over LGPS.

More generally, this approach may be applied beyond the Li-B-S system. Narrow electrochemical stability in fast ion conducting materials is a persistent issue. In this work, we explore a new approach to enhancing electrochemical stability by combining fast ion conducting

phases from the same chemical family. This approach could be extended to other solid electrolyte systems where multiple fast ion conducting phases exist and electrochemical stability is a concern.

Our study indicates that solid Li-B-S electrolytes may offer comparable or significantly improved performance over the best existing solid electrolytes in terms of ionic conductivity, electrochemical stability, materials cost, and weight. Synthesis routes for $\text{Li}_5\text{B}_7\text{S}_{13}$ and $\text{Li}_9\text{B}_{19}\text{S}_{33}$ has been reported.^[18] We welcome further computational and experimental investigations to assess the feasibility of synthesis as well as the long-term performance and stability of solid Li-B-S electrolytes.

Methods

DFT molecular dynamics simulation

For the simulation of DFT-MD in $\text{Li}_5\text{B}_7\text{S}_{13}$, $\text{Li}_2\text{B}_2\text{S}_5$, Li_3BS_3 , and $\text{Li}_9\text{B}_{19}\text{S}_{33}$, we utilize the Vienna Ab Initio Simulation Package (VASP)^[51] with the generalized gradient approximation (GGA) of Perdew-Burke-Ernzerhof (PBE)^[27] and the projector augmented wave (PAW) method.^[28] Our simulations use the NVT canonical ensemble. The unit cells for $\text{Li}_5\text{B}_7\text{S}_{13}$, $\text{Li}_2\text{B}_2\text{S}_5$, and Li_3BS_3 are obtained from the Materials Project database.^[52] We confirm the structures for $\text{Li}_2\text{B}_2\text{S}_5$ and Li_3BS_3 are the same as those originally reported in the literature, while the symmetry of $\text{Li}_5\text{B}_7\text{S}_{13}$ (space group Cc) is slightly different from that originally reported (space group C2/c).^[18] The unit cell for $\text{Li}_9\text{B}_{19}\text{S}_{33}$ is obtained from experimental reports^[18] via the Inorganic Crystal Structure Database, and a lattice relaxation was performed before initiating DFT-MD. We simulate supercells of 99, 71, 111, and 244 atoms, respectively. The large unit cell

for $\text{Li}_9\text{B}_{19}\text{S}_{33}$ caused DFT-MD simulations to run extremely slowly, so we ran two separate DFT-MD simulations in order to generate sufficient diffusivity data, rather than one long simulation. One lithium vacancy is introduced per unit cell. We use the Li_{sv} , B , and S pseudopotentials, a plane wave cutoff energy of 499 eV for all structures, and a gamma-point only k -mesh. This corresponds to a reciprocal lattice k -point density of 0.59, 0.68, 0.48, and 0.25 nm^{-3} . The VASP input files are generated using the *pymatgen.io.vasp.sets* module of Pymatgen.^[53,54]

Ionic conductivity calculation

To compute ionic conductivity from DFT-MD, we first compute the three-dimensional lithium diffusivity as a function of temperature $D(T)$, i.e. the average of the diagonal entries of the diffusivity tensor, and then calculate ionic conductivity through the Einstein relation. We compute $D(T)$ from one-sixth of the slope of the mean squared displacement of all Li over time at a given simulation temperature, $MSD(T,t) = \frac{1}{N_{\text{Li}}} \sum_{i=1}^{N_{\text{Li}}} [r_i(T,t) - r_i(T,0)]^2$. The squared displacement is averaged over all N_{Li} Li atoms in the simulation.

$$D(T) = \frac{1}{6} \frac{\Delta MSD}{\Delta t} \quad (2)$$

There are natural thermal fluctuations the MSD curves, which may be exacerbated by the shorter simulation times employed here. We take the slope of the MSD curve as the median slope when considering all time origins up to 75% of the total simulation time, in 100 equally-spaced intervals. We compute the uncertainty of the diffusivities as the 25th and 75th percentiles of these distributions of slope values.

Plots of the simulated Li MSD versus time curves for all four materials at all simulation temperatures are provided in the Supporting Information, Figure S1. Additionally, we plot the

MSD of the sublattice and confirm that none of the materials melt during the simulations. The displacement of the center of mass of Li was recorded throughout the simulations and we confirmed there was no significant movement of the Li center of mass in any materials at any temperature.

The diffusivities and uncertainties are calculated at 900 K, 700 K, 400 K, and 293 K. The RT diffusivities are converted to ionic conductivities using the Einstein relation:

$$\sigma(T) = \frac{D(T)nq^2}{k_B T} \quad (3)$$

where n is the Li ion number density and q is the average effective charge on the Li ions. We compute the average charge on Li using the converged electron densities computed from DFT and the Bader charge analysis methods of Henkelman et al.^[55–58] In these DFT calculations we use a plane wave cutoff of 520 eV and a Monkhorst Pack k -point mesh with a density of at least 1000 per atom. We compute the Bader charge on $\text{Li}_5\text{B}_7\text{S}_{13}$, $\text{Li}_2\text{B}_2\text{S}_5$, and Li_3BS_3 to be 0.89, 0.90, and 0.88, respectively.

Low temperature simulations often do not provide sufficient statistics to make a converged prediction of diffusivity, so we assume Arrhenius dynamics to extrapolate the high temperature diffusivities to room temperature. In all cases, the simulations at 293 K are taken to be not converged, while the convergence at 400 K is clear for $\text{Li}_5\text{B}_7\text{S}_{13}$ but questionable for the other three phases. We note the work of He et al.^[30] that suggests considering diffusivities to be converged when the total Li MSD is on the order of $100 \text{ \AA}^2$, which rules out all 400 K simulations except that of $\text{Li}_5\text{B}_7\text{S}_{13}$. We compute the RT ionic conductivity with the 400 K results included in the extrapolation and without. Fig. 1 shows all computed conductivities but

the dotted extrapolation lines are fit only to the 900 K, 700 K and 400 K data for $\text{Li}_5\text{B}_7\text{S}_{13}$, and the 900 K and 700 K data for all other phases. We note that the 400 K data for $\text{Li}_9\text{B}_{19}\text{S}_{33}$ and 293 K data for $\text{Li}_5\text{B}_7\text{S}_{13}$ align closely to the extrapolation. For the other two phases, the simulated low temperature conductivities are significantly higher than the extrapolations.

To compute the uncertainties associated with the extrapolation of conductivity to RT, we fit a simple linear model along the upper and lower limits of the error bars (on log scale) versus inverse temperature for each composition. The predicted error bar width at inverse room temperature is assumed to be the difference in these two linear extrapolations at room temperature.

To confirm convergence in Li ionic conductivity with respect to unit cell size and Li vacancy concentration, we simulate DFT-MD in all three phases at 900K with doubled unit cells and zero Li vacancy concentration. For $\text{Li}_5\text{B}_7\text{S}_{13}$ and Li_3BS_3 , we find very close alignment between the Li diffusivity of these simulations with the original simulations: 0.88 vs. $1.0 \text{ \AA}^2 \text{ ps}^{-1}$ for $\text{Li}_5\text{B}_7\text{S}_{13}$ and 0.55 vs. $0.61 \text{ \AA}^2 \text{ ps}^{-1}$ for Li_3BS_3 . For $\text{Li}_2\text{B}_2\text{S}_5$, the doubled unit cell with zero Li vacancy concentration showed no Li diffusivity after 33 ps of simulation. We do not perform this analysis for $\text{Li}_9\text{B}_{19}\text{S}_{33}$ given the large unit cell size.

Convex hull calculation

To calculate the electrochemical stability windows provided in Figures 3 and Table 1, we compute the grand potential $\Phi(\mu_{\text{Li}})$ of all phases at a range of applied Li chemical potentials $0 \leq \mu_{\text{Li}} \leq 5 \text{ V vs. Li/Li}^+$ and construct the convex hull. The convex hull represents the lowest grand

potential that can be accessed at a given composition and this defines the thermodynamically stable phase(s).

For a grand potential phase diagram at a given chemical potential, we employ a modified Graham scan algorithm^[59] for drawing the convex hull. Beginning with the lowest grand potential end member on the left-hand side of the diagram, phase i , we compute the angle θ_{ij} from the horizontal formed when drawing a line from this phase to all other phases j . Defining θ_{ij} as zero along the positive y -axis and increasing counterclockwise, we search for the phase j^* that forms an angle closest to π provided that $\pi < \theta_{ij} < 2\pi$ (i.e. j^* is down and to the right). This phase j^* is identified as sitting on the convex hull and a line is drawn between the two phases. For any phases k existing above the convex hull between phases i and j^* , the energy above the hull is computed as the grand potential difference between this line and the grand potential of phase k . The same procedure is then repeated, with the search now spreading out from the new phase j^* . This is repeated until the search identifies the lowest potential end member on the right-hand side of the phase diagram.

Ternary phase diagram

The ternary Li-B-S phase diagram shown in Fig. 2 is calculated using the *pymatgen.analysis.phase_diagram* module of pymatgen.^{37,38}

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The authors have no financial or commercial Conflict of Interest.

ASSOCIATED CONTENT

Supporting Information available: Mean squared displacement (MSD) data from DFT-MD for $\text{Li}_5\text{B}_7\text{S}_{13}$, Li_3BS_3 , $\text{Li}_2\text{B}_2\text{S}_5$, and $\text{Li}_9\text{B}_{19}\text{S}_{33}$ at 900 K, 700 K, 400 K, and 293 K.

Figures and Tables

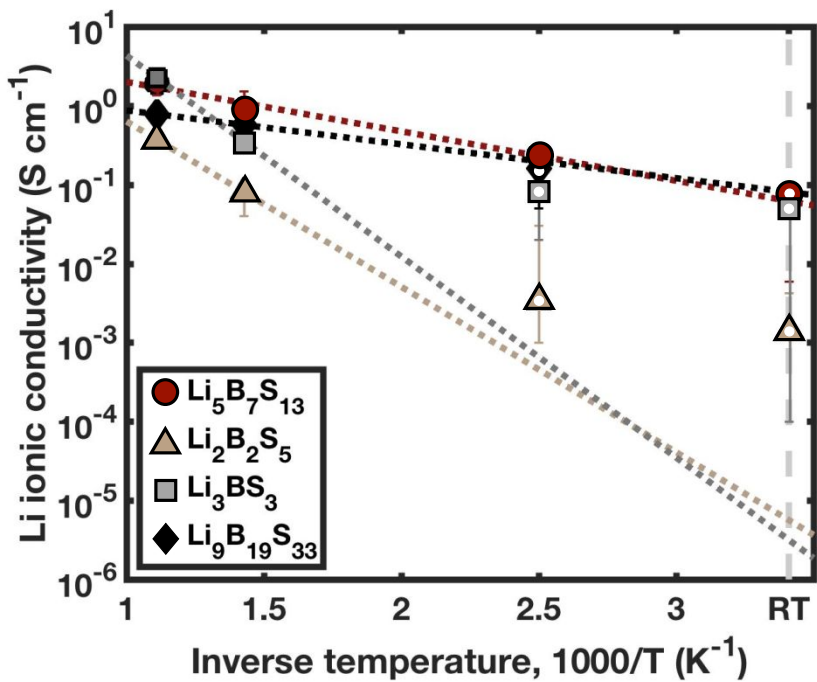


Figure 1: Temperature dependence on Li ionic conductivity for $\text{Li}_5\text{B}_7\text{S}_{13}$, $\text{Li}_2\text{B}_2\text{S}_5$, Li_3BS_3 , and $\text{Li}_9\text{B}_{19}\text{S}_{33}$. We simulate DFT MD at 900 K, 700 K, 400 K, and 293 K for four promising materials from the Li-B-S system, $\text{Li}_5\text{B}_7\text{S}_{13}$, $\text{Li}_2\text{B}_2\text{S}_5$, Li_3BS_3 , and $\text{Li}_9\text{B}_{19}\text{S}_{33}$. We compute the average Bader charge on Li and the slope of the Li mean squared displacement (MSD) curves then employ equations (2) and (3) to compute the Li ionic conductivity. These values represent the average of the diagonal elements of the conductivity tensor. The diffusivity is determined from the slope of the MSD curve. The error bars represent the 75th and 25th percentile of slopes in the diffusivity considering all time origins from 0 to 75% of the final run time in 100 equally-spaced intervals. If Arrhenius dynamics are assumed, the high temperature ionic conductivity extrapolates to give the following RT ionic conductivities: 74 mS cm⁻¹ for $\text{Li}_5\text{B}_7\text{S}_{13}$, 0.0056 mS cm⁻¹ for $\text{Li}_2\text{B}_2\text{S}_5$, 0.0031 mS cm⁻¹ for Li_3BS_3 , and 80 mS cm⁻¹ respectively. This would make $\text{Li}_5\text{B}_7\text{S}_{13}$ and $\text{Li}_9\text{B}_{19}\text{S}_{33}$ over five times more conductive than LGPS and two of the fastest solid Li ion conductors predicted to date. The markers with white dots at their centers denote simulations that did not reach an MSD of approximately 100 Å² over the simulation period; we assume these results to not be converged and leave them out of the Arrhenius extrapolation.³⁰ If the 400 K simulations for $\text{Li}_2\text{B}_2\text{S}_5$, Li_3BS_3 , and $\text{Li}_9\text{B}_{19}\text{S}_{33}$ are factored into the extrapolations, the RT ionic conductivity predictions become: 0.16 mS cm⁻¹ for $\text{Li}_2\text{B}_2\text{S}_5$, 9.7 mS cm⁻¹ for Li_3BS_3 , 56 mS cm⁻¹ for $\text{Li}_9\text{B}_{19}\text{S}_{33}$.

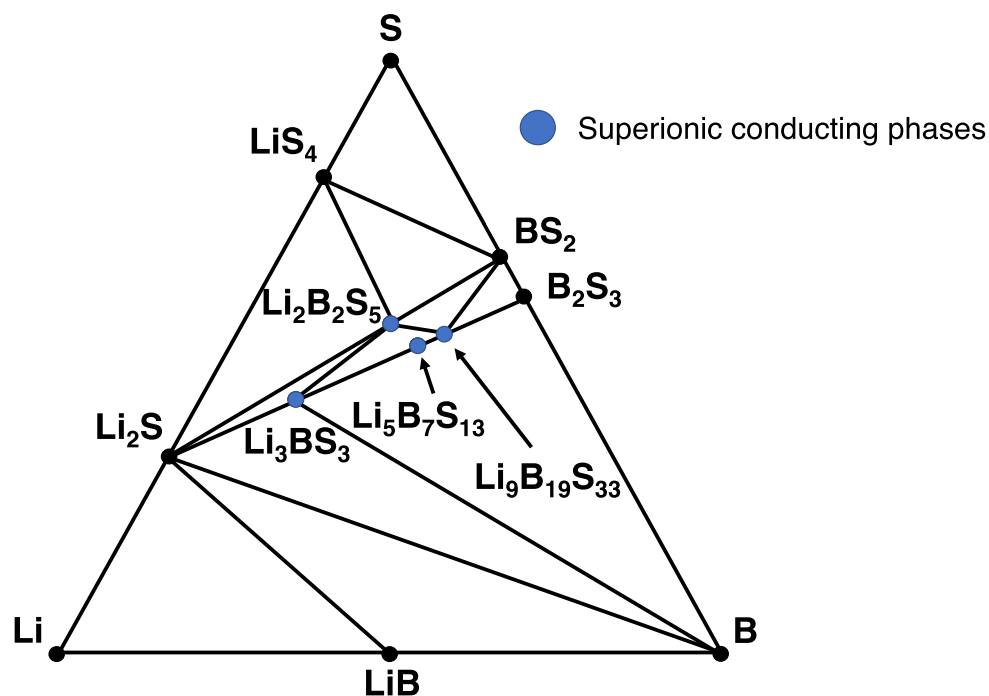


Figure 2: Ternary phase diagram of the Li-B-S system. We predict $\text{Li}_5\text{B}_7\text{S}_{13}$, $\text{Li}_2\text{B}_2\text{S}_5$, Li_3BS_3 , and $\text{Li}_9\text{B}_{19}\text{S}_{33}$ to be stable Li superionic conductors at room temperature. This suggests at least one of these four superionic phases will be present in many possible $\text{Li}_x\text{B}_y\text{S}_z$ compositions, suggesting the Li-B-S system may exhibit fast Li-ion conduction even when the material composition changes via interfacial degradation. The discovery of additional stable phases and their incorporation onto this phase diagram may evolve this conclusion.

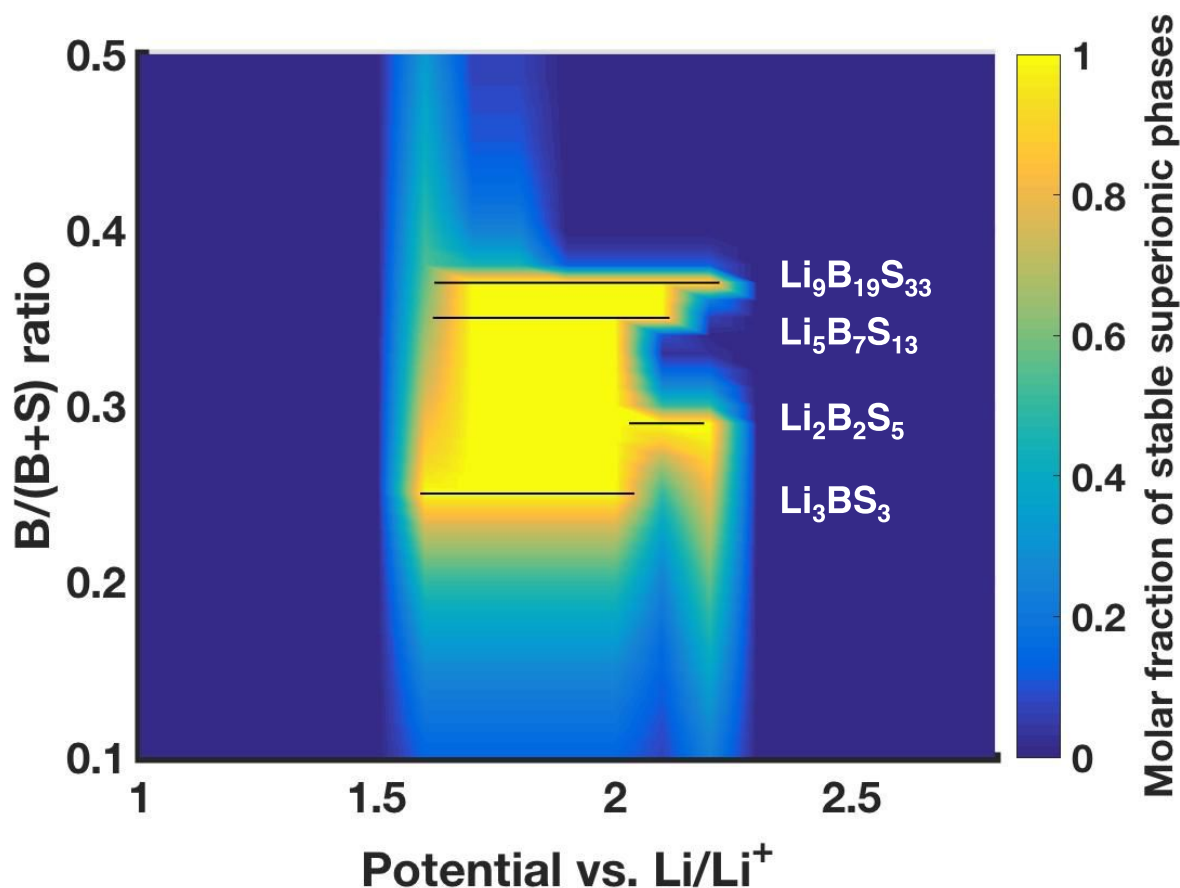


Figure 3: Superionic molar fraction versus composition and potential. The molar fraction of thermodynamically stable superionic phases is plotted as a function of the B/(B+S) ratio and electrode potential. The optimal B/(B+S) ratios are those which exhibit high molar fraction of superionic phases over the widest potential range. No superionic phases are predicted to be thermodynamically stable below approximately 1.6 V or above 2.2 V vs. Li/Li⁺, although high kinetic barriers may render them metastable. The B/(B+S) values associated with the pure phases are denoted with horizontal black lines, and the width of the black lines indicates the pure phase electrochemical stability window. The highest molar fraction of superionic phases over the widest potential range occurs around the B/(B+S) ratio of 0.365, (B:S ratio of 1:74). This ratio corresponds to the Li₉B₁₉S₃₃ pure phase.

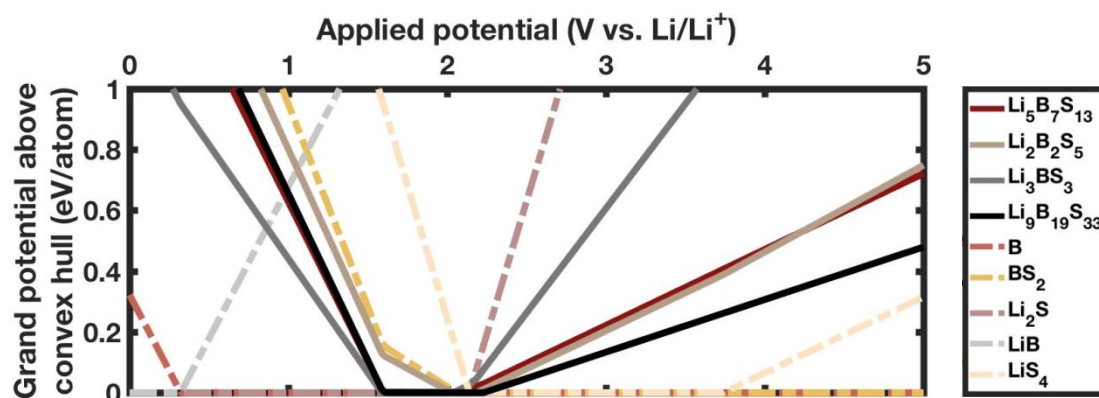


Figure 4: Analysis of electrochemical stability. We compute thermodynamic electrochemical stability windows for all stable phases in the Li-B-S system using PBE DFT. Single phases are predicted to be thermodynamically electrochemically stable at over the range of applied voltages for which they lie on the grand potential function convex hull. The thermodynamic electrochemical stability windows of the four superionic Li-B-S phases all lie between 1.59 and 2.22 V vs. Li/Li⁺. Outside of this voltage range, the difference between the single phase grand potential and the convex hull grows linearly. Although the kinetics behind the various Li-B-S degradation reactions are not known, we assume the reactions may be stabilized until the grand potential above the convex hull of the Li-B-S phases exceeds 0.4 eV/atom. Our calculations suggest Li₃BS₃ may be the most stable against reduction of all four superionic Li-B-S phases, possible remaining kinetically stabilized against anode potentials as low as 1 V. Li₉B₁₉S₃₃ appears to be the most stable against oxidation, and may remain stabilized against cathode potentials as high as 4.5 V.

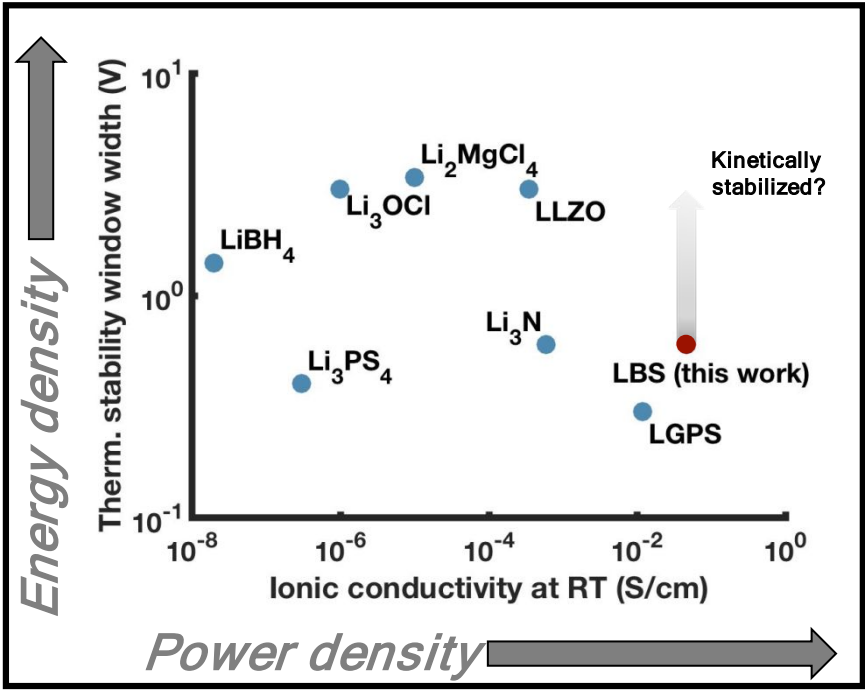


Figure 5: Solid electrolyte performance characteristics. Here we provide the thermodynamic electrochemical stability window widths computed with PBE DFT^[5,37,38] and experimentally reported ionic conductivities^[4,60–65] (on log-log scale) of several known solid Li-ion conductors, including the predicted values for the best Li-B-S electrolyte compositions reported here. The overall ionic conductivity of 46 mS cm⁻¹ in Li-B-S is computed by taking the weighted mean of the bulk ionic conductivities of Li₃B₇S₁₃, Li₂B₂S₅, Li₃BS₃, and Li₉B₁₉S₃₃, where the weights are proportional to the thermodynamic electrochemical stability window widths of the phases. Taking the ionic conductivity as an indicator of power density and the electrochemical stability as an indicator of energy density, we predict Li-B-S enables twice the energy density of LGPS and three times the power density. By comparison, LLZO exhibits a stability window five times wider than Li-B-S, but an ionic conductivity value over 100 times lower.

B molar ratio, $\xi = \mathbf{B}/(\mathbf{B} + \mathbf{S})$	Reactions at electrode potentials as indicated above arrows, with superionic phases in bold; $x = \frac{1}{\xi} - 1$
$0 \leq \xi \leq 0.25$	$x\text{Li}_2\text{S} + \text{LiB} \xrightarrow{0.36} x\text{Li}_2\text{S} + \text{B} \xrightarrow{1.59} \mathbf{Li}_3\mathbf{BS}_3 + (x-3)\text{Li}_2\text{S} \xrightarrow{2.04} 0.5\mathbf{Li}_2\mathbf{B}_2\mathbf{S}_5 + (x-2.5)\text{Li}_2\text{S} \xrightarrow{2.14} 0.5\mathbf{Li}_2\mathbf{B}_2\mathbf{S}_5 + (0.25x-0.63)\text{LiS}_4 \xrightarrow{2.19} (0.25x-0.5)\text{LiS}_4 + \text{BS}_2$
$0.25 \leq \xi \leq 0.286$	$x\text{Li}_2\text{S} + \text{LiB} \xrightarrow{0.36} x\text{Li}_2\text{S} + \text{B} \xrightarrow{1.59} 0.33x\mathbf{Li}_3\mathbf{BS}_3 + (1-0.33x)\text{B} \xrightarrow{1.62} (0.88x-1.63)\mathbf{Li}_3\mathbf{BS}_3 + (0.38-0.13x)\mathbf{Li}_5\mathbf{B}_7\mathbf{S}_{13} \xrightarrow{2.03} (2x-5)\mathbf{Li}_3\mathbf{BS}_3 + (3-x)\mathbf{B}_2\mathbf{S}_5 + (x-2.5)\text{Li}_2\text{S} \xrightarrow{2.14} 0.5\mathbf{Li}_2\mathbf{B}_2\mathbf{S}_5 + (0.25x-0.63)\text{LiS}_4 \xrightarrow{2.19} (0.25x-0.5)\text{LiS}_4 + \text{BS}_2 \xrightarrow{3.76} (x-2)\text{S} + \text{BS}_2$
$0.286 \leq \xi \leq 0.333$	$x\text{Li}_2\text{S} + \text{LiB} \xrightarrow{0.36} x\text{Li}_2\text{S} + \text{B} \xrightarrow{1.59} 0.33x\mathbf{Li}_3\mathbf{BS}_3 + (1-0.33x)\text{B} \xrightarrow{1.62} (0.88x-1.63)\mathbf{Li}_3\mathbf{BS}_3 + (0.38-0.13x)\mathbf{Li}_5\mathbf{B}_7\mathbf{S}_{13} \xrightarrow{2.03} (0.78x-1.44)\mathbf{Li}_2\mathbf{B}_2\mathbf{S}_5 + (x-2)\mathbf{B}_2\mathbf{S}_5 + (5-2x)\text{BS}_2 \xrightarrow{2.19} (0.25x-0.5)\text{LiS}_4 + \text{BS}_2 \xrightarrow{3.76} (x-2)\text{S} + \text{BS}_2$
$0.333 \leq \xi \leq 0.35$	$x\text{Li}_2\text{S} + \text{LiB} \xrightarrow{0.36} x\text{Li}_2\text{S} + \text{B} \xrightarrow{1.59} 0.33x\mathbf{Li}_3\mathbf{BS}_3 + (1-0.33x)\text{B} \xrightarrow{1.62} (0.88x-1.63)\mathbf{Li}_3\mathbf{BS}_3 + (0.38-0.13x)\mathbf{Li}_5\mathbf{B}_7\mathbf{S}_{13} \xrightarrow{2.03} (0.78x-1.44)\mathbf{Li}_2\mathbf{B}_2\mathbf{S}_5 + (2-x)\mathbf{Li}_5\mathbf{B}_7\mathbf{S}_{13} + (7x-13)\text{BS}_2 \xrightarrow{2.11} (0.4-0.2x)\mathbf{Li}_9\mathbf{B}_{19}\mathbf{S}_{33} + (3.8x-6.6)\text{BS}_2 \xrightarrow{2.22} (2-x)\text{B}_2\text{S}_3 + (2x-3)\text{BS}_2$
$0.35 \leq \xi \leq 0.365$	$x\text{Li}_2\text{S} + \text{LiB} \xrightarrow{0.36} x\text{Li}_2\text{S} + \text{B} \xrightarrow{1.59} 0.33x\mathbf{Li}_3\mathbf{BS}_3 + (1-0.33x)\text{B} \xrightarrow{1.62} 0.077x\mathbf{Li}_5\mathbf{B}_7\mathbf{S}_{13} + (1-0.54x)\text{B} \xrightarrow{1.62} (1.19x-2.06)\mathbf{Li}_5\mathbf{B}_7\mathbf{S}_{13} + (0.81-0.44x)\mathbf{Li}_9\mathbf{B}_{19}\mathbf{S}_{33} + (3.8x-6.6)\text{BS}_2 \xrightarrow{2.22} (2-x)\text{B}_2\text{S}_3 + (2x-3)\text{BS}_2$
$0.365 \leq \xi \leq 0.4$	$x\text{Li}_2\text{S} + \text{LiB} \xrightarrow{0.36} x\text{Li}_2\text{S} + \text{B} \xrightarrow{1.59} 0.33x\mathbf{Li}_3\mathbf{BS}_3 + (1-0.33x)\text{B} \xrightarrow{1.62} 0.077x\mathbf{Li}_5\mathbf{B}_7\mathbf{S}_{13} + (1-0.54x)\text{B} \xrightarrow{1.62} 0.03\mathbf{Li}_9\mathbf{B}_{19}\mathbf{S}_{33} + (1-0.58x)\text{B} \xrightarrow{1.87} (0.22x-0.03)\mathbf{Li}_9\mathbf{B}_{19}\mathbf{S}_{33} + (3.67-2.11x)\text{B}_2\text{S}_3 \xrightarrow{2.22} (2-x)\text{B}_2\text{S}_3 + (2x-3)\text{BS}_2$
$0.4 \leq \xi \leq 1$	$x\text{Li}_2\text{S} + \text{LiB} \xrightarrow{0.36} x\text{Li}_2\text{S} + \text{B} \xrightarrow{1.59} 0.33x\mathbf{Li}_3\mathbf{BS}_3 + (1-0.33x)\text{B} \xrightarrow{1.62} 0.077x\mathbf{Li}_5\mathbf{B}_7\mathbf{S}_{13} + (1-0.54x)\text{B} \xrightarrow{1.62} 0.03\mathbf{Li}_9\mathbf{B}_{19}\mathbf{S}_{33} + (1-0.58x)\text{B} \xrightarrow{1.87} 0.33\text{B}_2\text{S}_3$

Table 1: Thermodynamic reaction pathways. We provide the overall reaction for the Li-B-S electrolyte system as calculated with DFT grand potential methods, as a function of the B/(B+S) molar ratio, ξ , which runs from 0 (only Li and S) to 1 (only Li and B). The electrochemical potential (versus Li/Li⁺) at which each reaction is predicted to occur is provided above the reaction arrows, in order of increasing potential. The products on the far left of each set of reactions are predicted to be stable against Li metal (0 V) while the products on the far right are predicted to be stable against any potential above the potential listed over the furthest-right arrow. Superionic phases are denoted in bold

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TOC Graphic

